Chapter 22
Developing Countries: Growth, Crisis, and Reform

■ Chapter Organization

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Summary

■ Chapter Overview

This chapter provides the theoretical and historical background students need to understand the macroeconomic characteristics of developing countries, the problems these countries face, and some proposed solutions to these problems.

The chapter begins by discussing how the economies of developing countries differ from industrial economies. The wide differences in per capita income and life expectancy across different classes of all countries are striking. Some economic theories predict growth convergence, and there is evidence of such a pattern among industrialized nations, but no clear pattern emerges among developing countries. Some have grown rapidly, while others have struggled. Overall however, developing countries have accounted for an increasingly large portion of both global GDP and global economic growth.

There are important structural differences between developing economies and industrial economies. Governments in developing countries have a pervasive role in the economy, setting many prices and limiting transactions in a wide variety of markets; this can contribute to higher levels of corruption. These governments often finance their budget deficits through seigniorage, leading to high and persistent inflation. The economies of developing countries are typically not well diversified, with a small number of commodities providing the bulk of exports. These commodities, which may be natural resources or agricultural products have exhibited a volatile boom and bust cycle over the past two decades. Finally, economies of developing countries typically lack developed financial markets and often rely on fixed exchange rates and capital controls.

There is a discussion of the use of seigniorage in developing countries in the text. You may want to use the discussion of seigniorage in the text as a springboard for a more in-depth discussion of this topic. In particular, you could present a model of where seigniorage revenue is a function of the inflation rate chosen. The function is concave, at first increasing but eventually decreasing as high inflation leads people to hold less money (see Figure 22-1 below). It is much like the Laffer curve for taxation. This helps explain how similar seigniorage revenues may come from widely different inflation levels.

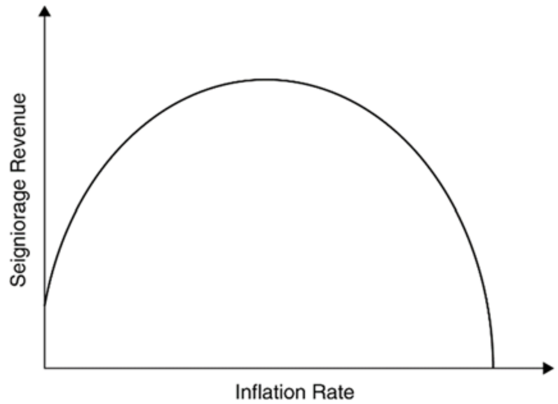

Figure 22-1

In principle, developing countries (and the banks lending to them) should enjoy large gains from intertemporal trade. Developing countries, with their rich investment opportunities relative to domestic saving, can build up their capital stocks through borrowing. They can then repay interest and principal out of the future output the capital generates. Developing-country borrowing can take the form of equity finance, direct foreign investment, or debt finance, including bond finance, bank loans, and official lending. These gains from intertemporal trade are threatened by the possibility of default by developing countries. Developing countries have defaulted in many situations over time, from 19th century American states to most developing countries in the Depression to the debt crisis in the 1980s. If lenders lose confidence, they may refuse further lending, forcing developing countries to bring their current account into balance. These crises are driven by similar self-fulfilling mechanisms as exchange rate crises or bank runs (and are often referred to as “sudden stops” when financial flows stop running to developing countries seemingly without warning), and the discussion of debt default provides an opportunity to revisit the ideas of currency crises and bank runs before a full-fledged discussion of the East Asian crisis.

It is important to recognize the different types of financing available to countries. Bond funding, bank borrowing, or official lending can all provide debt-oriented funding, while foreign direct investment or portfolio investment in firms can provide equity financing. In addition, countries can borrow in their own currency or in another currency. The chapter discusses the problem of “original sin” where many countries are unable to borrow in their own currency due to both problems in global capital markets and countries’ own histories of poor economic policies.

The next section of the chapter focuses on the experiences of Latin America. In the 1970s, inflation became a widespread problem in Latin America, and many countries tried using a *tablita*, or crawling peg. The strategy, though, did not stop inflation, and large real appreciations were the result. Government-guaranteed loans were widespread, leading to moral hazard. By the early 1980s, collapsing commodity prices, a rising dollar, and high U.S. interest rates precipitated default in Mexico followed by other developing countries.

After the debt crisis stretched through most of the decade and slowed developing country growth in many regions, debt renegotiations finally loosened burdens on many countries by the early 1990s.

After the debt crisis appeared to be ending, capital began to flow back into many developing countries. These countries were finally undertaking serious economic reform to stabilize their economy. The chapter details these efforts in Argentina, Brazil, Chile, and Mexico and also discusses how crisis unfortunately returned to some of these countries. For example, Argentina settled with its foreign creditors in 2005 for around a third of their debt obligations. This led to a brief period of stability, but in 2020 Argentina again reached a point where it needed to restructure its foreign debt following a default in 2019. Efforts to reduce corruption and implement economic reforms in Brazil and Mexico have been halting and limited by political cosiderations.

Next, the chapter covers the success and subsequent crisis in Asia. The causes of success, such as high savings, strong education, stable macroeconomics, and high levels of trade are considered. Some aspects of the economies that remained weak, such as low productivity growth and weak financial regulation are also discussed. The crisis, beginning in August 1997, is explained in detail along with its spread to other developing countries. The lessons of these years of growth and crisis are summarized as: choosing the right exchange rate regime, the importance of banking, proper sequencing of reforms, and the importance of contagion.

Because developing countries have relatively small financial markets, they are more susceptible to spillover effects from events in rich country financial markets. The monetary policy of the United States in particular has significant impacts on developing country financial markets. Empirical evidence suggests a strong correlation between the global financial cycle and economic growth in developing countries. When asset (equities, bonds, commodities, etc.) prices are high, economic growth in developing countries is strong. Similarly, a crash in global asset prices triggers a slowdown in developing country growth.

Given the economic impact of the global financial cycle on developing country growth, the open economy trilemma may actually collapse to a dilemma for developing countries. They can either pursue independent monetary policy with strict capital controls or allow for capital mobility and accept the economic influence of the global financial cycle. Note that developing countries choice of exchange rate regime appears not to matter since even a country with a flexible exchange rate will have limited ability to affect economic growth through monetary policy given the outsized influence of global financial markets. That said, there is some evidence that developing countries with flexible exchange rates experience less volatility in economic growth as a result of global financial shocks as the exchange rate can still act to dampen these shocks.

Finally, the chapter concludes with a section on current debates in the growth literature, chiefly a discussion of the relative importance of geography and institutions in driving income growth and levels. These factors are crucial to understanding the "Lucas puzzle" of why capital does not appear to be flowing to developing countries despite their low levels of capital, suggesting a high return in these countries. Rather, it appears that capital is flowing from the developing world to rich countries! Possible explanations for this puzzle include lower levels of human capital in developing countries as well as weaker institutions.

■ Answers to Textbook Problems

1. The amount of seigniorage governments collect does not grow monotonically with the rate of monetary expansion. The real revenue from seigniorage equals the money growth rate times the real balances held by the public. But higher monetary growth leads to higher expected future inflation and (through the Fisher effect) to higher nominal interest rates. To the extent that higher monetary growth raises the nominal interest rate and reduces the real balances people are willing to hold, it leads to a fall in real seigniorage. Across long-run equilibriums in which the nominal interest equals a constant real interest rate plus the monetary growth rate, a rise in the latter raises real seigniorage revenue only if the elasticity of real money demand with respect to the expected inflation rate is greater than −1. Economists believe that at very high inflation rates this elasticity becomes very negative (quite large in absolute value).
2. As discussed in the answer to Problem 1, the real revenue from seigniorage equals the money growth rate times the real balances held by the public. Higher monetary growth leads to higher expected future inflation, higher nominal interest rates, and a reduction in the real balances people are willing to hold. In a year in which inflation is 100% and rising, the amount of real balances people are willing to hold is less than in a year in which inflation is 100% and falling; thus, seigniorage revenues would be higher in 1990, when inflation is falling, than in 2000, when inflation was rising.
3. Although Brazil's inflation rate averaged 147% between 1980 and 1985, its seigniorage revenues, as a percentage of output, were less than half the seigniorage revenues of Sierra Leone, which had an average inflation rate of 43%. Because seigniorage is the product of inflation and real balances held by the public, the difference in seigniorage revenues reflects lower holdings of real balances in Brazil than in Sierra Leone. In the face of higher inflation, Brazilians find it more advantageous than residents of Sierra Leone to economize on their money holdings. This may be reflected in a financial structure in which money need not be held for very long to make transactions due to innovations such as automatic teller machines.
4. Under interest parity, the nominal interest rate of the country with the crawling peg will exceed the foreign interest rate by 10% because expected currency depreciation (equal to 10%) must equal the interest differential. If the crawling peg is not fully credible, the interest differential will be higher as the possibility of a large devaluation makes the expected depreciation larger than the announced 10%.

5. Capital flight exacerbates debt problems because the government is left holding a greater external debt itself but may be unable to identify and tax the people who bought the central bank reserves that are the counterpart of the debt and now hold the money in foreign bank accounts. To service its higher debt, therefore, the government must tax those who did not benefit from the opportunity to move funds out of the country. There is thus a change in the domestic income distribution in favor of people who are likely to be quite well off already. Such a regressive change may trigger political problems.
6. There may have been less lending available to private firms than to state-owned firms if lenders felt that state guarantees ensured repayment by state-owned firms. (In some cases, such as that of Chile, however, the government was pressured *ex post* into taking over the debts even of private borrowers.) Private firms may also have faced more discipline from the market—their operating losses are unlikely to be covered with public revenues. Private firms would therefore have had to restrict borrowing to investment projects of high quality.
7. By making the economy more open to trade and to trade disruption, liberalization is likely to enhance a developing country's ability to borrow abroad. In effect, the penalty for default is increased. In addition, of course, a higher export level reassures prospective lenders about the country's ability to service its debts in the future. Finally, by choosing policies that international lenders consider sound, such as open markets, countries improve lenders assessment of their creditworthiness.
8. Cutting investment today will lead to a loss of output tomorrow, so this may be a very shortsighted strategy. Political expediency, however, makes it easier to cut investment than consumption.
9. If Argentina dollarizes its economy, it will buy dollars from the United States with goods, services, and assets. This is, in essence, giving the U.S. Federal Reserve assets for green paper to use as domestic currency. Because Argentina already operates a currency board holding U.S. bonds as its assets, dollarization would not be as radical as it would be for a country whose central banks hold domestic assets. Argentina can trade the U.S. bonds it holds for dollars to use as currency. When money demand increases, the currency board cannot simply print pesos and exchange them for goods and services; it must sell pesos and buy U.S. government bonds. So, in switching to dollarization, the government has not surrendered its power to tax its own people through seigniorage; it already does not have that power.

   Still, though, through dollarization, Argentina loses interest by holding non-interest-bearing dollar bills instead of interest-bearing U.S. Treasury bonds. Thus, the size of the seigniorage given to the United States each year would be the lost interest (the U.S. nominal interest rate times the money stock of Argentina). This comes on top of the fact that any expansion of the money supply requires sending real goods, services, or assets to the United States for dollars (as they do with bonds under the currency board). This is not a long-run loss because Argentina could cash in those dollars (just as it could the bonds) for goods and services from the United States whenever it wants. So, what they lose is the interest they should be getting every year they hold the dollars.

10. No. Looking simply at countries that are currently industrialized and finding convergence is not a valid way to test convergence. Countries that are currently well off may have started from a variety of circumstances, but by only choosing countries that are currently wealthy, we are forcing a finding of convergence. It was an understandable error because currently rich countries are ones that are far more likely to have sufficiently long data series to test convergence, but only by looking at a broad array of countries can we truly test whether we should expect those countries that are currently poor to "catch up" to their richer counterparts.
11. The moral hazard comes from the fact that borrowers may borrow in a foreign currency, assuming the government will keep its promise to hold the exchange rate constant. Rather than hedging against the risk of exchange rate volatility, these borrowers assume the government will prevent the risk from materializing. The moral hazard comes from the fact that these borrowers engage in a risky behavior and assume the government will keep the exchange rate fixed—in part because of the promise to do so, but in part to prevent damage to these firms who have borrowed in a foreign currency.
12. Liability dollarization means that not only do participants in global financial markets face exchange rate risk, but many citizens simply participating in local markets face these risks. This means that a depreciation can have far more widespread effects throughout the economy. If mortgages are in U.S. dollars (as in Argentina in 2001), then workers who earn the local currency will suddenly be unable to pay their mortgage as the local currency value of their debt payments go up, but their income does not. This can lead to widespread defaults throughout the economy unless something is done to alleviate the impact. One of the only choices, though, is to convert the dollar loans to local currency loans, but this has a serious impact on the lenders if they have balanced their dollar assets with dollar liabilities. Finally, if enough local borrowers go bankrupt due to the adverse balance sheet effects, this could cause enough defaults to make the whole banking system insolvent.
13. The production function in both countries is given by $Y = AK^{\alpha}L^{1-\alpha}$. We can convert this into a per worker production function by dividing both sides by L. $Y/L = AK^{\alpha}L^{-\alpha} = A(K/L)^{\alpha}$. For ease of notation, let lowercase letters denote per worker values so that $y = Ak^{\alpha}$.

    From Table 22-2, we see that for 2017, y_{Ind} = \$6,548 and y_{US} = \$54,586. Thus, the ratio of the two countries out per worker is $y_{\text{Ind}}/y_{\text{US}}$ = \$6,548/\$54,586 = 0.12. Output per worker in India is 12% as large as output per worker in the United States.

Given these numbers, what value for A (multifactor productivity) would set the two countries marginal products of capital equal to one another? We can find this by looking at the MPK function: $MPK = \alpha AK^{\alpha-1} L^{1-\alpha} = \alpha Ak^{\alpha-1}$

Comparing the MPK to y, we can see that MPK = $\alpha y/k$. If we take the ratio of the MPK's for India and the United States, we will get $MPK_{\text{Ind}}/MPK_{\text{US}} = (y_{\text{Ind}}/y_{\text{US}})(k_{\text{US}}/k_{\text{Ind}})$. If the MPK's are equal between the two countries, then the ratio will be equal to 1. Thus, we know that for the MPK's to be equal, it must be the case that $y_{\text{Ind}}/y_{\text{US}} = k_{\text{Ind}}/k_{\text{US}}$.

We can show that $k = (y/A)^{1/\alpha}$. Plugging into the expression above gives $y_{\text{Ind}}/y_{\text{US}} = [(y_{\text{Ind}}/A_{\text{Ind}})/(y_{\text{US}}/A_{\text{US}})]^{1/\alpha}$. Solving for $A_{\text{Ind}}/A_{\text{US}}$ yields:

$A_{\text{Ind}}/A_{\text{US}} = (y_{\text{Ind}}/y_{\text{US}})^{1-\alpha} = 0.12^{1-\alpha}$

For example, if $\alpha = 0.3$ (capital's share of output is 30%), then the ratio $A_{\text{Ind}}/A_{\text{US}}$ must be equal to 0.227. For both countries to have the same marginal products of capital (and thus the same return on investment), multifactor productivity in India must be significantly lower (22.7% as large) than in the United States given the much higher level of capital in the United States.